03 · On-chain, privacy & the Solana moat

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> *Confidential DD dossier. Status tags:* ***BUILT+PROVEN / DESIGNED / RESEARCH-DIRECTION****.*
> *Companion to* `ECONOMICS-cooperative-commons.md` *(the token + reward economics).*

Why on-chain at all

Aiko's energy-based design has one honest cost: **it grows storage as it learns.** Every action trains a prototype and folds new history; run it long enough on one laptop and the weights + ledger accrue. The design turns that liability into a feature by **putting the weights on-chain**:

- **Weights become shareable.** A weight is just content-addressed ledger data (`cas.zig` — content-addressed store, dedup-for-free, integrity-checked; **BUILT+PROVEN**). Anyone with access can reconstruct the same prior from the same rows. You can *call your AI* — hand it to another device, fork it, audit it.
- **Storage cost is offloaded.** The machine holds a **frecency cache** — hot, frequently/recently used weights persist locally; cold ones live on-chain and are pulled on demand. Store **primarily on-chain**, cache locally what's in frequent use. **DESIGNED** (the caching policy; `cache.zig` is the local layer; the on-chain tier uses the anchoring path below).
- **Shared data stays cheap.** Content-addressing means a weight referenced by many contracts is stored (and paid for) **once** — the dedup is a property of the addressing, not a special case. **BUILT+PROVEN** (`cas.zig`).

Status of full on-chain weight storage: DESIGNED, partially proven. The primitives are real — content addressing (`cas`), live Solana reads (`solana_rpc`, devnet-proven), on-chain hash anchoring (`iq_ledger` on `main`, anchors claim hashes to a deployed Solana program), and a rent model that bounds what goes on-chain (≤ ~4 KB on-chain, larger = commitment-only + off-chain). The end-to-end "weights live on-chain, laptop is a cache" loop is assembled from these but not yet running in production at scale.

The Solana moat

The strategic bet is distribution + storage, stated plainly:

> *Once the EBM approach is shown to work, frontier labs will copy it — it's proprietary tech, but it's copyable.* ***The moat is not the algorithm; it's the ledger.*** *No one in AI has a distributed ledger the size of Solana. A ledger that can hold all the weights in the world — and a distribution channel already in millions of wallets — is the durable asset. Win the distribution and the storage substrate before the technology is commoditized.*

And the economics close on themselves: **increased Solana transaction volume means the product pays for itself.** Every weight write, every attestation, every settlement is on-chain activity; as usage grows, the on-chain volume it generates funds the substrate it runs on. (The full reward economics — usage fees paid out to stakers in USDC, FDRY as the security stake, reward-not-slash — are in `ECONOMICS-cooperative-commons.md`, **BUILT+PROVEN locally**, with the real mainnet FDRY vault integration in `voltr.zig`, **BUILT+PROVEN**.)

Honest status of the moat: this is a *go-to-market and defensibility thesis*, not a technical lock-in that exists today. Solana gives distribution + cheap durable storage; it does not by itself prevent a well-funded lab from building the same harness on their own infrastructure. The moat compounds only if Aiko wins the network (the commons of shared, corroborated weights) first. See `05-RISK-honest-wall.md`.

Privacy — what holds, and what is a bet

The pillar is *"your information stays yours,"* and it decomposes into three distinct claims with three different maturities. Diligence should not conflate them:

- **Local inference + sealed data — BUILT/DESIGNED.** Inference runs *on your machine* over your own data plus a shared index. You don't ship your data to train someone else's model; you fire locally and contribute **signed, capability-gated verdicts** — attestations of *what worked*, not the content. Your data at rest is

sealed under your key (envelope encryption + the **receptor grant** access boundary — `capability.zig`'s opt-in receptor is **BUILT+PROVEN**: value is adopted only under a signed grant, measured at 100% cross-lineage enforcement).

- **Zero-knowledge credential binding — RESEARCH-DIRECTION (not built).** The pitch says "ZK-proof credential binding creates privacy you can verify." Be precise for diligence: **there is no production zero-knowledge system today.** The current on-chain identity/attestation layer uses hash commitments and signed lineage, *not* a ZK proof system. Real ZK credential binding (e.g. Pedersen/Ristretto commitments + Schnorr + KVAC, avoiding SNARK trusted setup) is designed on paper and is the intended path — but it is unbuilt. Treat "ZK" as a **direction**, not a shipped feature.
- **Query privacy over a shared index — UNSOLVED.** Your data *at rest* is private. But *querying* a shared index can leak what you're asking. True query privacy needs PIR / local replicas / ZK — a separate, hard problem, not solved by encryption-at-rest. **"Private" holds for your data; not yet for your attention.** (Stated identically in the foundation THESIS — this is not new caution, it's the standing honest wall.)

The one-line summary for diligence

On-chain is **real in its primitives** (content-addressing, live reads, anchoring, a real FDRY vault, the reward loop) and **a designed system** at the "weights fully live on-chain, laptop is a frecency cache" level. Privacy is **real for data-at-rest**, **a direction for ZK**, and **an open problem for query privacy**. The moat is a **distribution/storage thesis**, compelling but not a technical lock-in yet. Everything here is labelled so the seams are visible.